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MILITARY SPECIFICATION

CONSOLE, INPUT OUTPUT, MK 77 MODS 0, 1 AND 2

This specification has been approved by the Naval Ordnance Systems Command, Deparment of the Navy.

1. SCOPE

1.1 Scope. - This document establishes the requirements for Input Output Console Mk 77 Mods 0, 1 and 2.

1.2 Specification type. - This document is considered to be similar to a Type A, Form 1b specification as set forth in MIL-S-83490.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and publications of the issue in effect on the date of the invitation for bids form a part of this specification to the extent specified herein.

FSC 1220

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SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-S-901	Shock Test, H.I.(High-Impact), Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-A-5092	Adhesive, Rubber (Synthetic and Reclaimed Rubber Base).
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for.
MIL-G-12803	Gasket Material, Non-Metallic.
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation and Delivery of.
MIL-F-18870	Fire Control Equipment, Naval Ship and Shore, General Specification for.
MIL-I-26860	Indicator, Humidity, Plug, Color Change.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-109	Quality Assurance Terms and Definitions.

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MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-167	Mechanical Vibrations for Shipboard Equipment.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-461	Electromagnetic Interference Characteristics; Requirements for Equipment.
MIL-STD-470	Maintainability Program Requirements (For Systems and Equipments).
MIL-STD-781	Reliability Tests Exponential Distribution.
MIL-STD-1399	Interface Standard for Shipboard Systems, Electric Power, Alternating Current.

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND

IL 2652601	Index List for Input Output Console MK 77 Mod 0.
IL 2809001	Index List for Input Output Console MK 77 Mod 1.
IL 2807182	Index List for Input Output Console MK 77 Mod 2.

PUBLICATIONS

NAVAL ORDNANCE SYSTEMS COMMAND

QAP 13905A	Quality Assurance Provisions for Input Output Console MK 77 Mod 0, 1 and 2.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 General characteristics. - Input Output Console Mk 77 Mod 0, Mod 1, and Mod 2, which may hereinafter be referred to as simply the IOC, shall have punched tape, keyboard and printer communication with Digital Computer Mk 152. The IOC shall comprise the features indicated in Table I.

3.2 Construction. - Unless otherwise specified herein, the IOC shall be constructed utilizing parts, materials and processes in accordance with drawings and other referenced documents of the Index Lists IL 2652601, IL 2807182 and IL 2809001. Any construction features not so specified shall be in accordance with the requirements of MIL-F-18870. The IOC shall be deck mounted.

3.2.1 Degree of enclosure. - The IOC enclosure shall be drip-proof as specified in MIL-STD-108.

3.2.2 Size. - The IOC shall be separable into sections which shall pass through an opening 26 inches by 45 inches having rounded corners of 8 inch radius and through a hatch 30 inches square with rounded corners of 7.5 inch radius.

The IOC shall have no greater than the following dimensions:

	Mod 0	Mods 1 & 2
Height (of cabinet)	48 inches (plus 13.75 inches for manual tape rewinder)	63.3 inches (includes printer)
Width (of cabinet)	22.5 inches	22.5 inches
Depth (of cabinet)	24.5 inches	24.5 inches
Desk height (without printer)	29 inches	N/A
Desk width	46 inches	N/A
Desk depth	24.5 inches	N/A

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Table I. Input Output Console Mk 77 configurations

Feature	Input Output Console Mk 77		
	Mod 0	Mod 1	Mod 2
Punch Tape Reader	Yes	Yes	Yes
Punch Tape Perforator	Yes	Yes	Yes
Manual Punch Tape Winder	Yes	Yes	Yes
Keyboard	Yes	Yes	Yes
Printer	Yes	Yes	Yes
Free-standing Deck	Yes	No	No
Type of Interface	Slow	Slow	Fast
Computer Interface	One	One	One
RF Shielding	Yes	Yes	No

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3.2.3 Weight. - Input Output Console Mk 77 Mod 0 shall not weigh more than 515 pounds, Input Output Console Mk 77 Mods 1 and 2 shall not weigh more than 405 pounds each.

3.2.4 Power sources. - The IOC shall operate as specified herein when supplied with 60 Hertz (Hz) and 400 Hz, Type I power as set forth in MIL-STD-1399, Section 103, qualified as follows and with consumption as indicated:

Logic Power

a)	Voltage	115
b)	Frequency	400 Hz
c)	Phase	3
d)	Power	250 watts

Motor Power

a)	Voltage	115
b)	Frequency	60 Hz
c)	Phase	1
d)	Power	600 watts

3.2.5 Cooling. - The IOC shall be air cooled. It shall generate heat at a rate which shall be no greater than 48 BTU per minute.

3.2.6 Physical interface. - The input-output channels of the IOC shall interconnect with an input-output channel of the computer by means of a pair of cables. The input data and associated control signals shall be on one cable, and the output data and associated control signals shall be on the other cable. The data and control signal distribution of the 90 pin connector shall be as specified in the appropriate drawing of the Index List.

3.2.7 Cabling and wiring. - All cabling and wiring shall be in accordance with MIL-F-18870. All cable connectors shall be mechanically keyed.

3.2.8 Grounding. - All grounds within a section or subassembly shall be connected to a common ground point on the IOC. Circuit grounding shall be in accordance with MIL-F-18870.

3.2.9 Finish and color. - Finish and color shall be in accordance with the documents in the Index List.

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3.2.10 Identification plates. - Identification plates shall be in accordance with MIL-F-18870.

3.3 Functional requirements. - The IOC shall have offline and on-line functions and shall perform as specified herein. The IOC shall function under manual control (off-line) as follows:

a) Print only data from punch tape.

b) Punch only data from punch tape.

c) Simultaneously print and punch data from punch tape.

d) Print only data entered at keyboard.

e) Simultaneously print and punch data entered at keyboard.

The IOC shall function under computer control (on-line) as follows:

a) Send to computer data from punch tape.

b) Simultaneously print and send to computer data entered at keyboard.

c) Punch only data received from computer.

d) Print only data received from computer.

e) Simultaneously punch and print data received from computer.

f) Print and send to computer data entered at keyboard and simultaneously punch data (same or different) returned from computer.

g) Send to computer data from punch tape and simultaneously print data (same or different) returned from computer.

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h) Send to computer data from punch tape and simultaneously punch data (same or different) returned from computer.

i) Send to computer data from punch tape and simultaneously both punch and print data (same or different) returned from computer.

3.3.1 Punch tape perforator. - The IOC shall perforate chad-type tape of 11/16, 7/8, and 1 inch widths with 5, 6, 7, or 8 channels. The IOC shall perforate dry paper, oiled paper, Mylar and metallized Mylar tape. The IOC shall perforate tape at a rate of up to 110 frames per second.

3.3.2 Punch tape reader. - The IOC shall read chad-type tape of 11/16, 7/8, and 1 inch widths with 5, 6, 7, or 8 channels. The IOC shall read dry paper, oiled paper, Mylar and metallized Mylar tape. The IOC shall read at a rate of up to 300 frames per second.

3.3.3 Keyboard-printer assembly. - When the keyboard function is enabled, the keyboard-printer shall generate each of the codes listed in Table II when the corresponding key or keys listed is operated. When the printer function is enabled, the keyboard-printer shall print-out the corresponding character listed in Table II for each of codes 0100001 through 1011010 (whether received from the computer or directly from the keyboard) and the following:

a) [for simultaneous operation of SHIFT and K keys.

b) \ for simultaneous operation of SHIFT and L keys.

c)] for simultaneous operation of SHIFT and M keys.

d) Blank character space for operation of space bar.

Whether or not specifically selected, the printer function shall print any data entered at the keyboard when the keyboard function is selected. The data entry keys of the keyboard-printer shall be interlocked to prevent operation of more than one key at a time. The keyboard-printer shall print 10 characters per inch and 6 lines per inch. It shall print a 72 character line.

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Table II. Keyboard-Printer Codes

Code (non-print)	Key(s) that generate code	Code (non-print)	Key(s) that generate code
000 0000	CTRL & @	001 0001	CTRL & Q
000 0001	CTRL & A	001 0010	CTRL & TAPE ON
000 0010	CTRL & B	001 0011	CTRL & X OFF
000 0011	CTRL & C	001 0100	CTRL & TAPE OFF
000 0100	CTRL & EOT	001 0101	CTRL & U
000 0101	CTRL & WRU	001 0110	CTRL & V
000 0110	CTRL & RU	001 0111	CTRL & W
000 0111	CTRL & BELL	001 1000	CTRL & X
000 1000	CTRL & H	001 1001	CTRL & Y
000 1001	CTRL & I	001 1010	CTRL & Z
000 1010	LINE FEED	001 1011	CTRL, SHIFT & K
000 1011	CTRL & K	001 1100	CTRL, SHIFT & L
000 1100	CTRL & L	001 1101	CTRL, SHIFT & M
000 1101	RETURN	001 1110	CTRL, SHIFT & ↑
000 1110	CTRL & N	001 1111	CTRL, SHIFT & ←
000 1111	CTRL & O	111 1101	ALT. MODE
001 0000	CTRL & P	111 1111	RUB OUT

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Table II. Keyboard-Printer Codes (Cont.)

Code (Printed)	Key(s) that generate code	Code (Printed)	Key(s) that generate code
010 0000	SPACE BAR	011 0001	1
010 0001	SHIFT & !	011 0010	2
010 0010	SHIFT & "	011 0011	3
010 0011	SHIFT & #	011 0100	4
010 0100	SHIFT & $	011 0101	5
010 0101	SHIFT & %	011 0110	6
010 0110	SHIFT & &	011 0111	7
010 0111	SHIFT & '	011 1000	8
010 1000	SHIFT & (	011 1001	9
010 1001	SHIFT &)	011 1010	:
010 1010	SHIFT & *	011 1011	;
010 1011	SHIFT & +	011 1100	SHIFT & <
010 1100	,	011 1101	SHIFT & =
010 1101	-	011 1110	SHIFT & >
010 1110	.	011 1111	SHIFT & ?
010 1111	/	100 0000	SHIFT & @
011 0000	0	100 0001	A

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Table II. Keyboard-Printer Codes (Cont.)

Code (Printed)	Key(s) that generate code	Code (Printed)	Key(s) that generate code
100 0010	B	101 0001	Q
100 0011	C	101 0010	R
100 0100	D	101 0011	S
100 0101	E	101 0100	T
100 0110	F	101 0101	U
100 0111	G	101 0110	V
100 1000	H	101 0111	W
100 1001	I	101 1000	X
100 1010	J	101 1001	Y
100 1011	K	101 1010	Z
100 1100	L	101 1011	SHIFT & K
100 1101	M	101 1100	SHIFT & L
100 1110	N	101 1101	SHIFT & M
100 1111	O	101 1110	SHIFT & ↑
101 0000	P	101 1111	SHIFT & ←

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3.4 Manual controls and indicators. - The IOC shall have the the following groups of controls and indicators:

a) Power, temperature alarm, and tape low indicators and controls located on the hood of the basic unit.

b) Functional controls located behind a door on the basic unit.

c) A power switch located on the punch tape reader on the basic unit.

d) Keyboard-printer data and function entry keys, controls and indicators located on the keyboard-printer assembly.

Except for data and function entry keys on the keyboard-printer assembly, all these controls and indicators shall operate as specified in Table III.

Table III. Controls and Indicators

Location	Label on Panel	Type of Device	Function
Hood of basic unit	PERFORATOR TAPE LOW	Green indicator	Lights when less than 5 to 10% of full roll of tape remains on perforator supply reel
Hood of basic unit	POWER LOGIC	Green indicator	Lights while power is being applied to logic power supplies
Hood of basic unit	POWER BLOWERS	Green indicator	Lights while power is being applied to blowers

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Table III. Controls and Indicators (Cont.)

Location	Label on Panel	Type of Device	Function
Hood of basic unit	POWER ON OFF	Two position momentary contact switch	Applies and removes power to power supplies and blowers
Hood of basic unit	OVER TEMP	Red indicator	115° (+0°, -5°) OVER TEMP lights; 140° (+0°, -5°)F all power goes off
Hood of basic unit	ALARM BYPASS NORMAL	Two position switch and indicator	NORMAL position alarm sounds at 115° (+0°, -5°)F. BYPASS position: no alarm, indicator lights.
Behind door of logic chassis	MC	Pushbutton switch	Stops console operation, and puts all logic in an initial state*
Behind door of logic chassis	OUTPUT	8-bit indicator-switch group	Each lights when operated* or while the corresponding output register is set under program control
Behind door of logic chassis	INT	Indicator-switch	Enables generation of the keyboard interrupt to the computer as specified in 3.5.1.2
Behind door of logic chassis	TAPE LEVELS	3 position rotary selector	Enables reading of selected tape channels (5, 6-7, or 8)

* Function disabled when OFF LINE/ON LINE switch is in ON LINE position.

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Table III. Controls and Indicators (Cont.)

Location	Label on Panel	Type of Device	Function
Behind door of logic chassis	FUNCTION	5 indicator-switch, clear switch pairs	When operated, each indicator-switch lights and enables corresponding function* (READ, KEYBOARD, PUNCH, PRINT, COPY): CLR (clear) switch extinguishes indicator and disables corresponding function*
Behind door of logic chassis	TAPE FEED	Indicator-switch	Lights and feeds tape (punching only sprocket holes) while operated*
Behind door of logic chassis	START READ CLR	Indicator-switch and pushbutton switch	When operated, indicator-switch lights and enables read operation*; CLR switch extinguishes light and disables read operation*
Behind door of logic chassis	READ READ ONE	Two position momentary contact switch	While operated to READ position, Console will read (if read function is enabled)*; when operated to READ ONE position, Console will read one frame (if read function is enabled) advance tape to next frame and stop*

* Function disabled when OFF LINE/ON LINE switch is in ON LINE position.

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Table III. Controls and Indicators (Cont.)

Location	Label on Panel	Type of Device	Function
Behind door of logic chassis	OFF LINE ON LINE	Two position momentary contact switch	ON LINE position: enables on-line Console operation as specified in 3.3. OFF LINE position: enables off-line Console operation as specified in 3.3
Punch tape reader	POWER ON	Two position switch	Controls power to punch tape reader motor
Keyboard-printer	LOC CR	Two position switch	Causes carriage return when operated
Keyboard-printer	LOC LF	Two position switch	Causes continuous line feed action while operated.
Keyboard-printer	REPT	Two position switch	Causes repetitious generation of code corresponding to data entry key operated.
Behind door of logic chassis	INPUT	8-bit indicator-switch group	Each lights when operated or while the corresponding input register bit is set under program control

* Function disabled when OFF LINE/ON LINE switch is in ON LINE position.

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3.5 Functional interface requirements. - The input-output channel of the IOC shall function as specified herein. The IOC shall recognize each binary one received on any control line by detecting the change in voltage level of the signal. It shall not recognize another binary one on that control line until its circuitry has been reset by a change in the voltage level of the signal in the opposite direction (a binary zero).

3.5.1 Input communication. - When interconnected as shown in Figure 1, the IOC shall transfer data to the computer as specified herein. Input channel control lines shall function as specified in Table IV.

3.5.1.1 Transfer of input data. - Whenever an Input Data buffer has been established for a channel, the IOC shall transfer data as follows:

a) The IOC shall place a word of data on the Input Data Lines.

b) The IOC shall set the Input Data Request Line (to indicate that a word of data is on the Input Data Lines).

c) In accordance with internal priority, the computer will detect the setting of the Input Data Request Line.

d) The computer will read the data word which is on the Input Data Lines.

e) The computer will set the Input Acknowledge Line (indicating that it has read the data word on the Input Data Lines).

f) The IOC shall detect the setting of the Input Acknowledge Line. (The IOC may clear the Input Data Request Line any time after detecting the setting of the Input Acknowledge line, but it shall clear the Input Data Request Line before the computer will recognize the next Input Data Request.)

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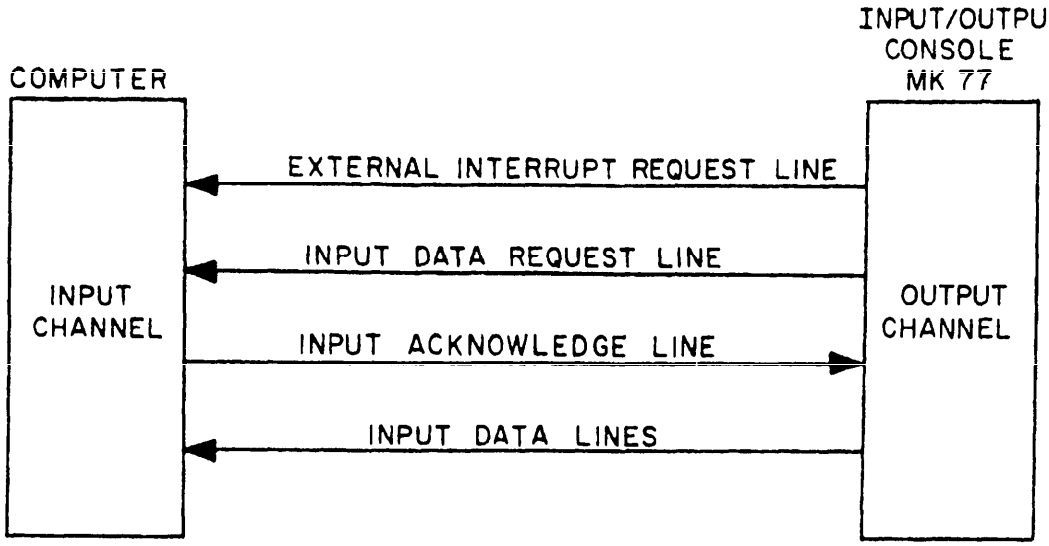


Figure 1. Input Communication Interface

Table IV. Function of Input Channel Control Lines

Name of Line	Direction of Signal	Function
Input Data Request Line	IOC to computer	Set condition shall indicate that the IOC has placed a word of data available to the computer on the Input Data Lines of that channel
Input Acknowledge Line	Computer to IOC	Set condition will indicate that the computer has read the Input Data Lines of that channel
External Interrupt Request Line	IOC to computer	Set condition shall indicate that the IOC has placed an Interrupt Code Word available to the computer on the Input Data Lines of that channel

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g) The computer will clear the Input Acknowledge Line before it reads the next word on the Input Data Lines.

The IOC shall repeat this sequence for each successive word of data until the block of data words specified by the Input Buffer Control Words has been transferred.

3.5.1.2 Transfer of an External Interrupt Code Word. - The IOC shall transfer an External Interrupt Code Word as follows:

a) The IOC shall place an External Interrupt Code Word on the Input Data Lines

b) The IOC shall set the External Interrupt Request Line.

c) The computer will detect the setting of the External Interrupt Request Line.

d) In accordance with internal priority, the computer will read the External Interrupt Code word on the Input Data Lines.

e) The computer will set the Input Acknowledge Line.

f) The IOC shall detect the setting of the Input Acknowledge Line. (The IOC may clear the External Interrupt Request Line any time after detecting the setting of the Input Acknowledge Line, but it shall clear the External Interrupt Request Line before the computer will recognize the next External Interrupt Request.)

g) The computer will clear the Input Acknowledge Line before it reads the next word on the Input Data Lines.

3.5.2 Output communication. - When interconnected as shown in Figure 2, the IOC shall accept data from the computer as specified herein. Output channel control lines shall function as specified in Table V.

3.5.2.1 Transfer of an External Function Code Word. - The IOC shall accept an External Function Code Word as follows:

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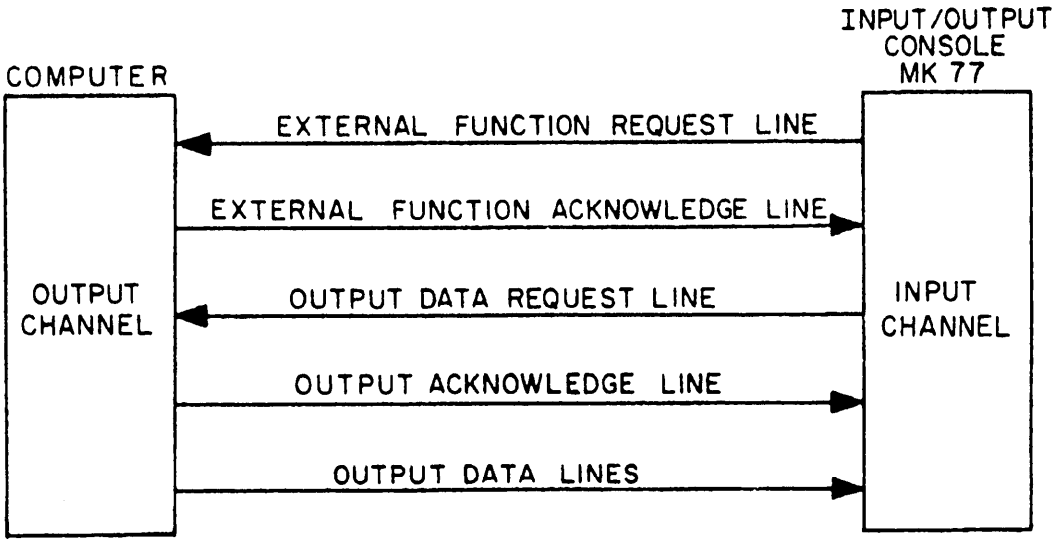


Figure 2. Output Communication Interface

Table V. Function of Output Channel Control Lines

Name of Line	Direction of signal	Function
External Function Request	IOC to computer	Set condition shall indicate readiness of the IOC to accept an External Function Code Word on that channel
External Function Acknowledge	Computer to IOC	Set condition will indicate that the computer has placed an External Function Code Word on the Output Data Lines of that channel
Output Data Request Line	IOC to computer	Set condition shall indicate readiness of the IOC to accept a word of data on that channel
Output Acknowledge Line	Computer to IOC	Set condition will indicate that the computer has placed a word of data on the Output Data Lines of that channel

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a) When the IOC is ready to accept an External Function Code Word, it shall set the External Function Request Line.

b) In accordance with internal priorities, the computer will detect the setting of the External Request Line.

c) The computer will place an External Function Code Word on the Output Data Lines.

d) The computer will set the External Function Acknowledge Line (to indicate that the External Function Code Word is on the Output Data Lines).

e) The IOC shall detect the setting of the External Function Acknowledge Line. (The IOC may clear the External Function Request Line any time after detecting the setting of the External Function Acknowledge Line, but it shall clear the External Function Request Line before the computer will recognize the next External Function Request.)

f) The IOC shall read the External Function Code Word which is on the Output Data Lines.

g) The computer will clear the External Function Acknowledge Line before it places the next word on the Output Data Lines.

3.5.2.2 Transfer of Output Data. - Whenever an Output Data buffer has been established for a channel, the IOC shall accept data as follows:

a) When the IOC is ready to accept data, it shall set the Output Data Request Line. (This may already have happened before the Output Data buffer was established)

b) In accordance with internal priority, the computer will detect the setting of the Output Data Request Line.

c) The computer will place a word of data on the Output Data Lines.

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d) The computer will set the Output Acknowledge Line (to indicate that a word of data is on the Output Data Lines).

e) The IOC shall detect the setting of the Output Acknowledge Line. (The IOC may clear the Output Data Request Line any time after detecting the setting of the Output Acknowledge Line, but it shall clear the Output Data Request Line before the computer will recognize the next Output Data Request.)

f) The IOC shall read the data word which is on the Output Data Lines.

g) The computer will clear the Output Acknowledge Line before it places the next word on the Output Data Lines.

The IOC shall repeat this sequence for each successive word of data until the block of data words specified by the Output Buffer Control Words has been transferred.

3.5.3 Electrical interface. - The slow interface (mod 0,1) shall be characterized by nominal values of 0 volts and -15 volts to represent a binary one and a binary zero, respectively, and by a switching threshold of -6.0 ±1.5 volts. Input Output Consoles Mk 77 Mod 0 and Mod 1 shall have slow interface. The fast interface shall be characterized by nominal values of 0 volts and -3 volts to represent a binary one and a binary zero respectively. Input Output Console Mk 77 Mod 2 shall have fast interface.

3.5.3.1 Input amplifiers (slow interface). - Each slow interface input amplifier circuit shall have the following characteristics:

a) The output of the circuit shall switch from a binary zero to a binary one whenever the input signal changes in the positive direction through the range of -7.5 volts to -4.5 volts.

b) The output of the circuit shall switch from a binary one to a binary zero whenever the input signal changes in the negative direction through the range of -4.5 volts to -7.5 volts.

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c) The output of the circuit shall not switch as a result of any input transient-pulse signal that has an integrated amplitude-duration of less than 15 volt-microseconds (a delay of 1.5 ±0.5 microseconds with a 15-volt step input).

d) The output of the circuit shall be a binary zero whenever its input is open-circuited.

e) The circuit shall not draw more than 4.0 milliamperes for a steady state binary one nor more than 1.0 milliamperes for a steady state binary zero.

f) The output of the circuit shall be a binary zero whenever the steady state input signal is more negative than -7.5 volts.

g) The output of the circuit shall be a binary one whenever the steady state input signal is more positive than -4.5 volts.

3.5.3.2 Line drivers (slow interface). - Each slow interface line driver circuit shall have the following characteristics when driving a line with any capacitance of up to 6000 picofarads. Each data line driver circuit that drives more than one line shall have proportionally more capability than specified herein.

a) The steady state output voltage representing a binary one shall be -1.5 volts to +1.5 volts.

b) The steady state output voltage representing a binary zero shall be -10.0 volts to -17.5 volts.

c) The voltage variation between all binary zero output signals on one channel shall not exceed 1.0 volt.

d) The voltage variation between all binary one output signals on one channel shall not exceed 1.0 volt.

e) The circuit shall be capable of supplying 4.0 milliamperes per line for a steady state binary one output.

f) The circuit shall be capable of supplying 1.0 milliampere per line for a steady state binary zero output.

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g) The output of the circuit shall switch in not more than 6.0 microseconds (measured between 10 and 90 percent amplitude points).

h) The output of the circuit shall switch at a rate of not more than 5.0 volts per microsecond.

i) The circuit shall be capable of driving the line and its termination (input amplifier) while switching as well as in the steady state condition.

j) The driver circuit shall present not less than 100,000 ohms impedance to the cable whenever power to a control line driver circuit is removed.

3.5.3.3 Input amplifier (fast interface). - Each fast interface input amplifier circuit shall have the following characteristics:

a) The output of the circuit shall switch from a binary zero to a binary one whenever the input signal changes in the positive direction through the range of -1.9 volts to -1.1 volts.

b) The output of the circuit shall switch from a binary one to a binary zero whenever the input signal changes in the negative direction through the range of -1.1 to -1.9 volts.

c) The output of the circuit shall not switch as a result of any input transient-pulse signal that has an amplitude of less than 7.5 volts if its duration and amplitude are common to both sides of the line (common mode).

d) The output of the circuit shall be a binary zero whenever its input in open-circuited.

e) The circuit shall not draw more than 1.5 milliamperes for a steady state binary one nor more than 0.5 milliampere for a steady state binary zero.

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f) The output of the circuit shall be a binary zero whenever the steady state input signal is more negative than -1.9 volts.

g) The output of the circuit shall be a binary one whenever the steady state input signal is more positive than -1.1 volts.

h) The input circuit shall present a terminal impedance to the line equivalent to a resistance of 150 to 180 ohms in series with a capacitance of 0.0068 to 0.0100 microfarad.

3.5.3.4 Line drivers (fast interface). - Each fast interface line driver circuit shall have the following characteristics when driving a line with any characteristic impedance between 120 ohms and 180 ohms:

a) The steady state output voltage representing a binary one shall be 0.0 volts to -0.5 volts.

b) The steady state output voltage representing a binary zero shall be -3.0 volts to -5.0 volts (it may be more negative if the input circuit being driven presents a more negative signal).

c) The circuit shall be capable of supplying 1.5 milliamperes per line for a steady state binary one output.

d) The circuit shall be capable of supplying 0.5 milliampere per line for a steady state binary zero output.

e) The output of the circuit shall switch in not more than 0.4 microseconds (measured between -0.5 volts and -3.0 volts).

f) The circuit shall be capable of driving the line(s) and termination(s) (input amplifier(s)) while switching as well as in the steady state condition.

g) Whenever power to a control line driver circuit is removed, it shall present not less than 100,000 ohms impedance to the cable.

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3.5.4 Timing. - Durations of signals and between signals in any communication sequence shall comply with the applicable limits specified in Figures 3 and 4. These limits are neither absolute nor necessarily typical, but rather they are minimums that denote the following dual requirement:

a) Neither the initiation nor the termination of any control or data signal shall occur sooner than specified in Figures 3 and 4.

b) The IOC shall be capable of recognizing data and control signals that occur at the times or at any time later than the times specified in Figures 3 and 4 and that exist for any duration equal to or greater than the durations specified in Figures 3 and 4.

3.5.5 Computer control. - Except for the keyboard interrupt sequence, all on-line data transfers shall be enabled by the following external function codes from the computer:

a) Code 0000011 enables printer output.

b) Code 0000101 enables perforator output.

c) Code 0000111 enables printer and perforator output.

d) Code 0011000 enables keyboard input.

e) Code 0101000 enables reader input*.

f) Code 1101000 enables start-read sequence.

g) Code 0011101 enables keyboard input and perforator output.

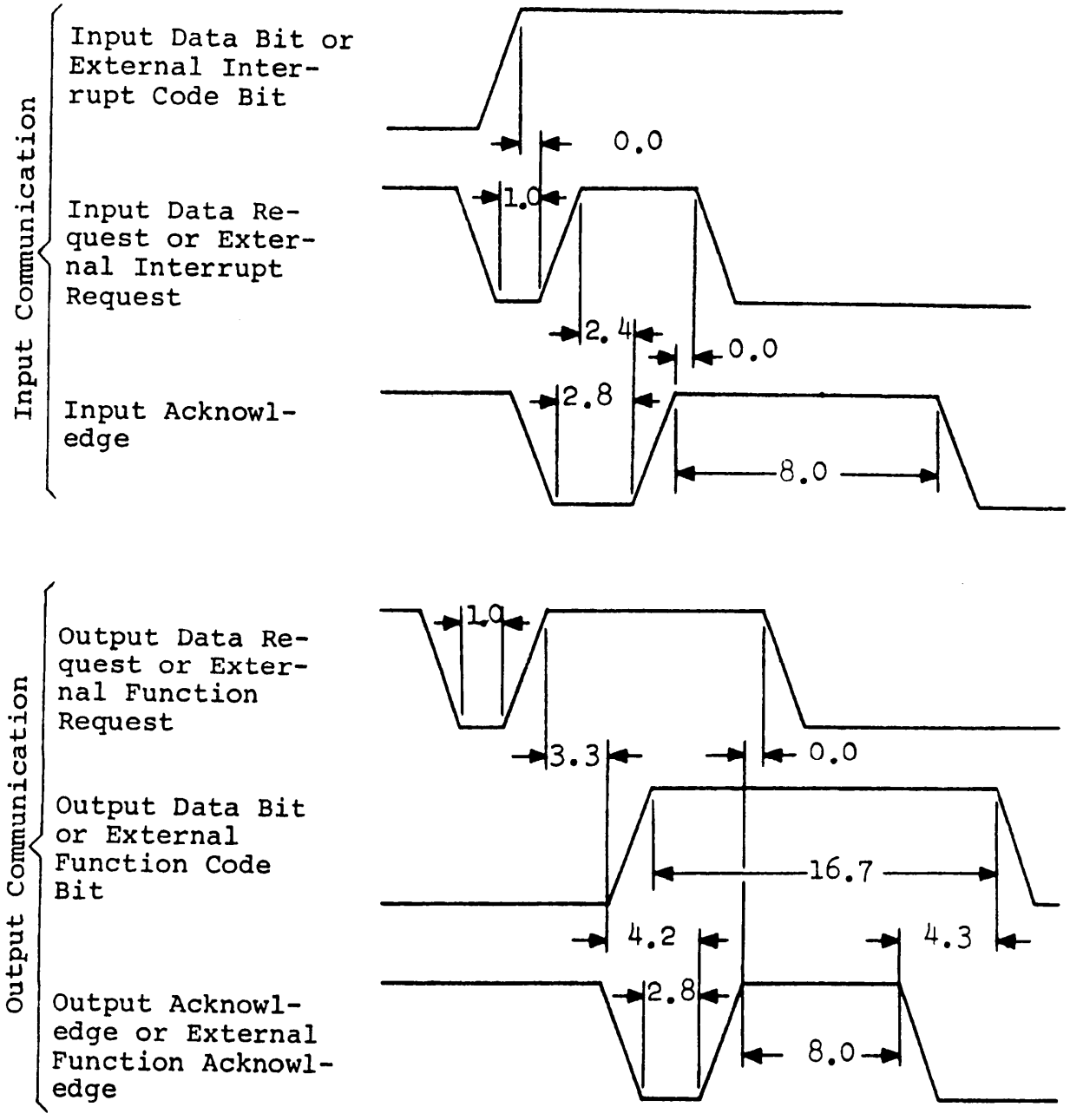
h) Code 0101011 enables reader input and printer output.

i) Code 0101101 enables reader input and perforator output*.

j) Code 0101111 enables reader input and printer and perforator output*.

*These codes must be used in conjunction with the start-read code (see 3.5.6).

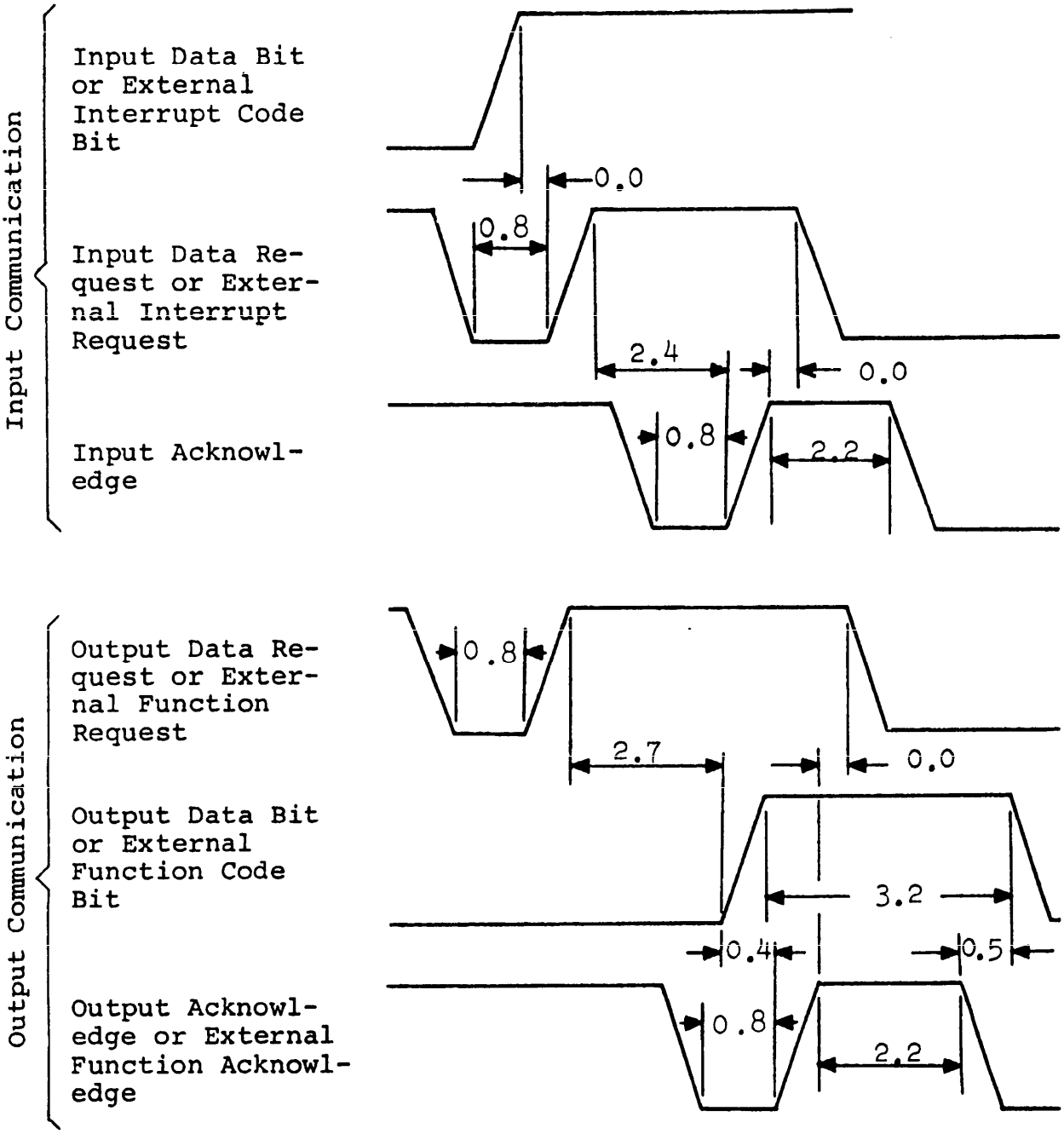
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NOTES: Upper level of each signal is a binary one; lower level is a binary zero. All times are in microseconds, are minimums, and are measured between points on the signal pulse that are either 10 percent or 90 percent of the amplitude shown. All transitions have a maximum duration of 6 microseconds and a maximum rate of 5 volts per microsecond.

Figure 3. Slow Interface Timing

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NOTES: Upper level of each signal is a binary one; lower level is a binary zero. All times are in microseconds, are minimums, and are measured between points on the signal pulse that are either -0.5 volt or -3.0 volts as shown. All transitions are maximum deviation of 0.4 microsecond.

Figure 4. Fast Interface Timing

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3.5.6 Detailed input-output sequences. -

3.5.6.1 Punch tape reader. - With tape loaded in the punch tape reader and the reader function enabled (see 3.5.5 and Table III), the IOC shall read tape by the following sequence upon receipt of the start-read external function code:

a) If a sprocket hole of the tape is positioned over the sensor, no advancement of the tape shall occur: if the tape is positioned so that the sensor is between sprocket holes, the clutch shall be engaged and the tape shall be advanced.

b) The next sprocket hole that reaches the sensor shall actuate the brake and the tape shall stop.

c) The signal caused by the data holes in each frame shall set or clear the corresponding input data lines through the action of the input register.

d) The signal caused by the sprocket hole shall cause the control circuitry to set the input-date request line.

e) Each frame is then transferred from the IOC to the computer by the transfer-of-input-data sequence as specified in 3.5.1.1.

3.5.6.2 Keyboard. - The keyboard input, in all modes, shall be printed by the printer. The keyboard input to the computer shall have two possible modes as follows:

a) When the keyboard Input Request-Acknowledge sequence is enabled, the IOC shall transfer each keyed character to the computer by the transfer-of-input-data sequence as specified in 3.5.1.1.

b) When the keyboard External Interrupt sequence is enabled (by a switch on the IOC, see Table III), the IOC shall transfer each keyed character to the computer by the External Interrupt Request transfer as specified in 3.5.1.2.

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3.5.6.3 Punch tape perforator. - When the punch tape perforator output function is enabled, the IOC shall transfer each frame to be punched from the computer by the transfer-of-Output-Data sequence specified in 3.5.2.2.

3.5.6.4 Printer. - When the printer function is enabled, the IOC shall transfer each character from the computer by the transfer-of-Output-Data sequence as specified in 3.5.2.2.

3.6 Reliability. - Each IOC shall have a mean-time-between-failure (MTBF) of not less than 200 hours.

3.7 Maintainability. - The mean-time-to-repair (MTTR) of the IOC shall not exceed 10 minutes. The MTTR includes diagnostic and active repair time but not logistic or administrative time.

3.7.1 Interchangeability. - All parts shall be interchangeable physically and electrically with corresponding parts without drilling, filing, or the use of undue force. The replacement part when installed shall not result in a departure from the performance specified herein. The IOC itself shall be interchangeable with the same type of IOC.

3.8 Environmental requirements. -

3.8.1 Ambient temperature. - The IOC shall operate, maintain performance as specified herein, and incur no physical damage within the ambient temperature range of 0 to 50° Celsius (C) (Class 4 in accordance with MIL-F-18870).

3.8.2 Storage temperature. - The IOC shall be capable of extended storage within the temperature range specified by MIL-F-18870, without physical damage or degradation of performance when restored to the operating temperature range.

3.8.3 Humidity. - The IOC shall operate satisfactorily within the humidity range specified by MIL-F-18870.

3.8.4 Fungus. - The IOC shall show no evidence of fungus growth after being subjected for extended periods to fungus and fungus supporting conditions.

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3.8.5 Corrosion resistance. - The IOC shall be corrosion resistant in accordance with MIL-F-18870.

3.8.6 Inclination. - The IOC shall be capable of normal operation when subjected to the pitch and roll conditions specified in MIL-F-18870.

3.8.7 Vibration. - The IOC shall be capable of meeting the Type I vibration requirements of MIL-STD-167.

3.8.8 Shock resistance. - The IOC shall be capable of meeting the Grade A, Class I shock requirements of MIL-F-18870.

3.8.9 Transport requirements. - The IOC shall meet the transport requirements of MIL-F-18870.

3.8.9.1 Air transport. - The IOC or components thereof shall not be damaged by air transportation at altitudes up to 50,000 feet at isometric pressures to 3.4 inches of mercury in non-pressurized compartments, in a non-operative condition.

3.9 Radio interference. - The IOC shall meet both the emission limitations and the susceptibility requirements for Type ID equipment as specified in MIL-STD-461. The emission limits shall apply for conditions of starting and stopping as well as for the normal operation of the equipment. This requirement shall not apply to Input Output Console Mk 77 Mod 2.

3.10 Mechanical noise. - The IOC shall meet the mechanical noise requirements of MIL-F-18870.

3.11 Insulation resistance. - Except for the printed wiring cards, the circuits of the IOC shall be capable of passing the insulation resistance tests specified herein.

3.12 Stability of adjustment. - The complete IOC shall operate as specified herein and shall not require readjustment under normal shipboard operating conditions except after repair or replacement. This requirement shall be included in any reliability program submitted by the supplier and shall be a minimum performance requirement for the equipment.

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3.13 Documentation. - All technical data and documentation shall be as specified in the contract or order.

3.14 Government loaned equipment. - Government loaned equipment shall be as specified in the contract or order.

3.15 Government furnished equipment. - Government furnished equipment shall be as specified in the contract or order.

3.16 Preproduction model. - Unless otherwise specified in the contract or purchase order, a preproduction model IOC shall be delivered to the inspection activity designated in the contract or purchase order for preproduction inspection. The preproduction model shall be manufactured using the materials, parts, processes, and procedures intended for production. Any production prior to approval of the preproduction model shall be at the supplier's risk.

3.17 Workmanship. - Workmanship shall be in accordance with Requirement 9 of MIL-STD-454. In addition, the standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercail laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. - Inspections (tests and examinations) shall be as defined in MIL-STD-109 and shall be classified as follows:

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a) Preproduction inspection.

b) Quality conformance inspection (acceptance inspection).

4.2.1 Measurements. - During all tests, the accuracy of instruments and test equipment used to control or monitor test parameters shall be verified periodically to the satisfaction of the procuring activity. All instruments and test equipment used in conducting the tests specified herein shall:

a) Conform to laboratory standards where calibration is traceable to the prime standards at the U.S. Bureau of Standards.

b) Unless otherwise specified herein, have an accuracy of at least one-tenth the tolerance for the variable to be measured.

4.2.2 Standard conditions. - Where applicable all quality conformance tests shall be conducted under standard conditions of ambient temperature, atmospheric pressure, and relative humidity. Standard conditions shall be as follows:

a)	Temperature	25 ±10 degrees C
b)	Relative humidity	90 percent or less
c)	Barometric pressure	28 to 32 inches of mercury

4.3 Sampling. -

4.3.1 Preproduction sample. - Unless otherwise specified in the contract or order, a preproduction sample of one IOC manufactured in accordance with the requirements of Section 3 of this specification shall be subjected to the preproduction tests detailed herein at an activity designated by the Naval Ordnance Systems Command. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the IOC by the supplier prior to the approval of the preproduction sample shall be at the supplier's risk.

4.3.2 Quality conformance sampling. - Each IOC offered for acceptance under contract shall be subjected to the examinations and tests specified in 4.5.1 and 4.5.4 through 4.5.8.

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In addition, samples selected from units having satisfactorily passed these inspections shall be subjected to the tests of 4.5.2 and 4.5.3 at an activity designated by the contracting officer. Unless otherwise specified in the contract or order, sampling for the tests of 4.5.2 and 4.5.3 shall be in accordance with MIL-STD-105, Inspection Level S-3.

4.4 Preproduction inspection. - Unless otherwise specified in the contract or order, the preproduction sample shall be subjected to the quality conformance inspections of 4.5.1 through 4.5.10. The preproduction sample may also be subjected to additional tests and examinations as determined by the Naval Ordnance Systems Command.

4.5 Quality conformance inspection. -

4.5.1 Examinations and tests. - The IOC shall be subjected to the examinations and tests specified in QAP 13095A and as otherwise specified herein. Failure to meet any of the inspection requirements shall be considered cause for rejection.

4.5.1.1 Packaging and packing inspection. - Inspection shall be conducted in accordance with MIL-E-17555 to assure compliance with the requirements of 5.1 and 5.2.

4.5.1.2 Marking inspection. - Examinations shall be conducted to ascertain that units, maintenance parts, and other items prepared for shipment are marked in accordance with the requirements of 5.3.

4.5.2 Degree of enclosure. - A drip-proof enclosure test shall be conducted in accordance with MIL-STD-108.

4.5.3 Environmental tests. - The IOC shall operate satisfactorily at the ambient conditions described below.

4.5.3.1 Nonoperating temperature test. - The IOC shall satisfactorily pass the nonoperating condition temperature test specified in MIL-F-18870.

4.5.3.2 Operating temperature test. - The IOC shall satisfactorily pass the operating temperature tests for Class 4 equipment as specified by MIL-F-18870.

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4.5.3.3 Humidity test. - The IOC shall satisfactorily pass the humidity tests as specified in MIL-F-18870.

4.5.3.4 Humidity and fungus. - The IOC shall be tested for resistance to humidity and fungus damage in accordance with MIL-E-5272. The IOC shall then pass the insulation resistance test of 4.5.4.

4.5.3.5 Salt spray. - In order to test for corrosion resistance, the IOC shall pass the salt spray test of MIL-F-18870.

4.5.3.6 Air transport. - The IOC shall be tested for its ability to withstand isometric pressures up to 3.4 inches of mercury (altitudes up to 50,000 feet) in a nonoperative condition, without damage.

4.5.3.7 Inclination test. - The IOC shall be tested for satisfactory operation while undergoing the inclination test in accordance with MIL-F-18870.

4.5.3.8 Vibration test. - The IOC shall be tested for satisfactory operation during and after the vibration test for Type I equipment in accordance with MIL-STD-167.

4.5.3.9 Shock test. - The entire IOC shall be subjected to a Type A, Medium Weight Shock test for Grade A, Class I equipment in accordance with MIL-S-901. Subassemblies and components shall be subjected to Type B or Type C (as appropriate) Light Weight Shock tests for Grade A, Class I equipment in accordance with MIL-S-901).

4.5.4 Insulation resistance test. - Each circuit shall be subjected to the insulation resistance tests specified in MIL-F-18870 except that test voltages shall not be applied to circuits which would incur damage. Test Condition A shall be utilized for low voltage circuits which will operate at a potential not over 25 volts and Test Condition B shall be used for all other circuits. All circuits shall show an insulation resistance of not less than 50 megohms.

4.5.5 Radio frequency noise interference. - Each IOC shall pass the tests for Class ID equipment as specified by MIL-STD-461.

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4.5.6 Mechanical noise test. - The IOC shall satisfactorily pass the mechanical noise test of MIL-F-18870.

4.5.7 Power system variations. - The IOC shall be tested for satisfactory operation under the power system variations contained in 3.2.4 of this specification. This test may be conducted during the performance of any or all of the tests of this specification at the discretion of the testing activity or as directed by the procuring activity.

4.5.8 Endurance. - After completion of all other tasks the IOC shall be tested for satisfactory operation. The IOC shall be operated and periodically tested for operation until a total of 100 hours operating time has elapsed and then shall again pass the test problems and otherwise operate satisfactorily. Test conditions during the endurance test may be any condition, or combination of conditions of inclination, ambient temperature, humidity and power system variations that are specified in the above test. Operating time shall be considered as time during which the IOC is actually performing its intended use. This test may be included as part of the reliability test specified herein.

4.5.9 Reliability. - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests as required by both the Qualification Phase and the Sampling Phase shall be conducted.

4.5.9.1 Qualification Phase. - Prior to the acceptance of equipments under the contract or order, a minimum of three IOC's shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of IOC's to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level C shall be used. The Accept-Reject Criteria for Test Plan XX shall be used.

4.5.9.2 Reliability Sampling Phase tests. - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the section entitled "Sampling Phase of Product on Reliability Tests". For the Sampling Phase, Test Level C shall be used. The Accept-Reject Criteria for Test Plan XXIV shall be used to determine the length of the tests (until an accept or reject decision is reached).

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4.5.9.2.1 Lot size for Sampling Phase. - The entire contract quantity (with the exception of those equipments used in the Qualification Phase) shall be considered one lot. The test program shall start the first month after the Qualification Phase has been completed. Samples shall be selected and placed on test in the same manner as though one month's production was a lot. The samples shall be tested until an accept or reject decision is reached. In the meantime, other equipments constructed shall be shipped. The test results shall be summarized monthly for the procuring activity. The procuring activity reserves the right to stop the acceptance of equipment at any time after one or more reject decisions have been reached, pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment quality control, etc., so that the entire contract quantity will show an accept decision.

4.5.9.3 Test details. - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted to and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.5.10 Maintainability. - Maintainability requirements shall be verified in accordance with MIL-STD-470.

4.6 Satisfactory operation defined. - Satisfactory operation or performance is hereby defined as the ability to pass the test problems established for acceptance tests set forth in QAP 13095A.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. -

5.1.1 Level A. -

5.1.1.1 Cleaning, drying and preservation application. - Not applicable.

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5.1.1.2 Unit packaging. - Unit packaging of the IOC shall be in accordance with Method IIA of MIL-E-17555. Each unit shall contain one section of the IOC individually packaged. Gasket material shall be in accordance with MIL-G-12803, Type I, cemented in place with adhesive conforming to MIL-A-5092, Type III. The humidity indicator shall be in accordance with MIL-I-26860, Type II. The rough handling tests of MIL-P-116 shall not apply.

5.1.1.3 Intermediate package. - Not applicable.

5.1.1.4 Cushioning. - The unit package and pack shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices. At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent. The pH range of the material shall be between 6.0 and 8.0.

5.1.2 Level C. - In addition to the following, packaging shall be in accordance with MIL-E-17555.

5.1.2.1 Cleaning, drying and preservation application. - Not applicable.

5.1.2.2 Unit packaging. - Unit packaging of the IOC shall be in accordance with Method II of MIL-P-116.

5.1.2.3 Intermediate package. - Not applicable.

5.2 Packing. -

5.2.1 Level A. -

5.2.1.1 Exterior containers. - The IOC packaged as described in 5.1.1.2 shall require no over-packing.

5.2.2 Level C. - The IOC packaged in accordance with 5.1.2.2 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking. - Shipping containers shall be marked in accordance with MIL-STD-129, including serial numbers.

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6. NOTES

6.1 Intended use. - Input Output Console Mk 77 will be used in conjunction with Digital Computer Mk 152 as an input output device.

6.2 Ordering data. - Procurement documents should include the following:

a) Title, identification number and date of this specification.

b) Documentation required, if any

c) Government loaned equipment

d) Government furnished equipment

e) Number of equipments to be subjected to the inspections of 4.5.2 and 4.5.3 (see 4.3.2).

f) Additional examinations to be performed on the preproduction sample (see 4.4).

g) Disposition of sample subjected to shock tests (see 4.5.3.9).

h) Level of shipment required (see section 5).

6.2.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in MIL-P-116.

6.2.2 Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

For Level A. - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level C. - This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

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6.3 Definitions. - For the purpose of this specification, the following definitions shall apply:

Input - Input to the computer.

Output - Output from the computer.

Set - To put into or to be in the logical one state (0 volts nominal on the Input-Output lines.)

Clear - To put into or be in the logical zero state (minus 15 volts nominal on the Input-Output lines for slow interface, minus 3 volts for fast interface.)

Frame - A lateral row of perforations in punch tape.

6.4 IOC Comparison. - Table VI provides a summary of the basic requirements of the IOC.

Custodian:
NAVY - OS

Preparing Activity
NAVY - OS
Project Number 1220-N271

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Table VI. Input Output Console Mk 77 comparison

	Mod 0	Mod 1	Mod 2
Index List	IL 2652601	IL 2809001	IL 2807182
Teletype Model 35 Keyboard-printer	Mounted on desk	Mounted on top of cabinet	Mounted on top of cabinet
Manual Tape Winder	---	Mounted on top of cabinet	Mounted on top of cabinet
Height (of cabinet)	48 inches (plus 13.75 in. for manual tape rewinder)	63.3 inches (including keyboard-printer)	63.3 inches (including keyboard-printer)
Width (of cabinet)	22.5 inches	22.5 inches	22.5 inches
Depth (of cabinet)	24.5 inches	24.5 inches	24.5 inches
Desk height (w/o printer)	29 inches	---	---
Desk width	46 inches	---	---
Desk depth	24.5 inches	---	---
Weight	515 lbs.	405 lbs.	405 lbs.
RF Shielding	Yes	Yes	No
Interface	Slow	Slow	Fast

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-82618(OS) "Console, Input Output', MK 77 MODS 0, 1 And 2

ORGANIZATION

CITY AND STATE	CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*	DATE

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